

## DNA

- **DNA** – Polymer of deoxyribonucleotides
- **Nucleoside** = Nitrogenous base + Pentose sugar (linked through *N* – **glycosidic bond**)  
Example – adenosine, deoxyadenosine, cytidine, etc.
- **Nucleotide** = Nucleoside + Phosphate group (linked through **phosphoester bond**)
- Many nucleotides link together through 3' – 5' **phosphodiester bond** to form polynucleotide chain (as in DNA and RNA).
- In course of formation of polynucleotide chain, a phosphate moiety remains free at 5' end of ribose sugar (5' end of polymer chain) and one -OH group remains free at 3' end of ribose (3' end of polymer chain).

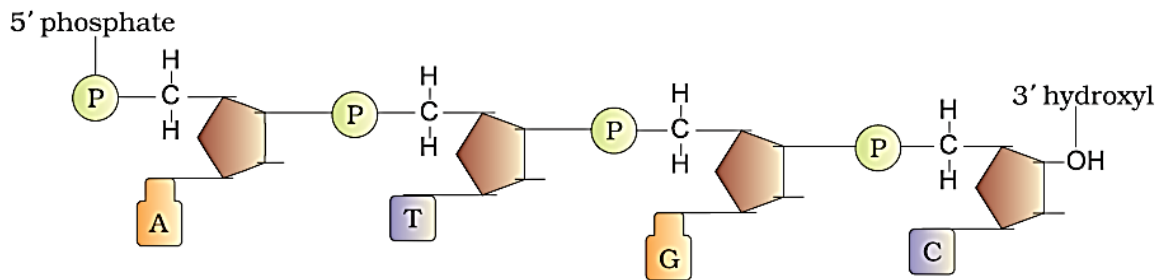


Figure: A polynucleotide chain

## Double Helix Model for the Structure of DNA

Scientists involved

- **Friedrich Meischer** – First identified DNA as an acidic substance present in nucleus and named it as 'Nuclein'
- **Wilkins and Franklin** – Produced X-ray diffraction data for DNA structure
- **Watson and Crick** – Proposed double helix structure model for DNA based on X-ray diffraction data
- **Erwin Chargaff** – Proposed that in ds DNA, ratio of A:T and C:G remain same and are equal to one

## The salient features of double helix structure of Watson & Crick model of DNA:

- DNA is made of two polynucleotide chains.
- The backbone of DNA strand composed of repeated units of sugar and phosphate molecules, and the bases project inside.

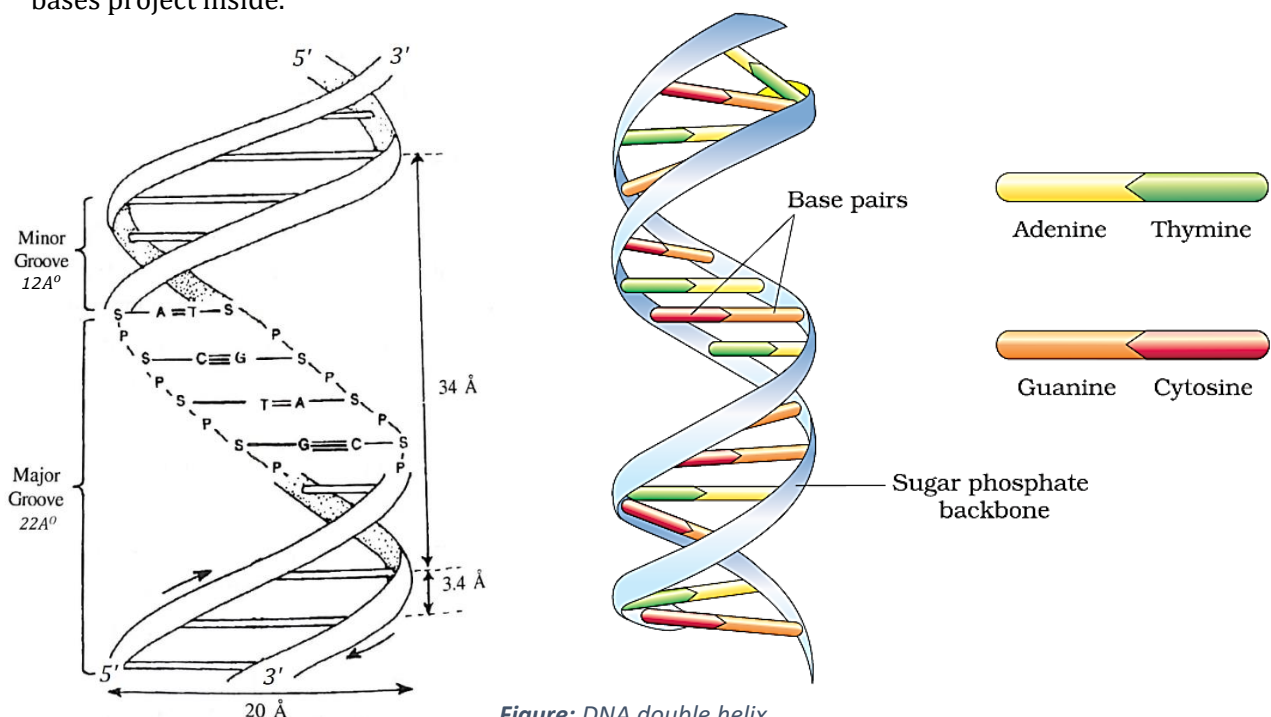


Figure: DNA double helix

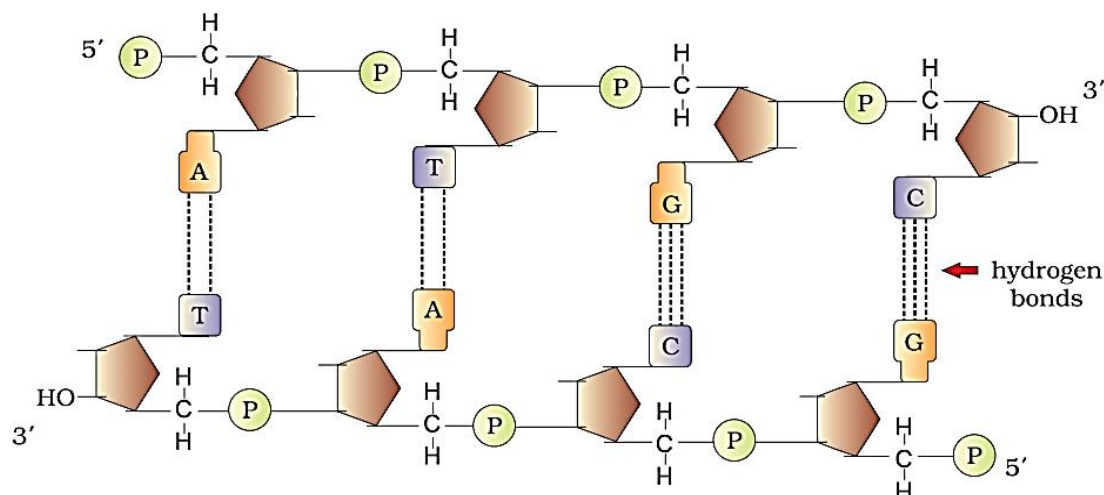
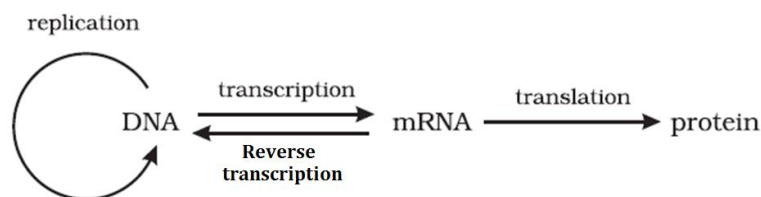


Figure: Double stranded polynucleotide chain

- The two chains have anti-parallel polarity. It means, if one chain has polarity  $5' \rightarrow 3'$ , the other has  $3' \rightarrow 5'$ .
- The pairing of nitrogen bases are always between a specific purine and specific pyrimidine i.e., between A and T, G and C vice versa. This type of base pairing is called **complementary base pairing**.
- The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). There are two hydrogen bonds between A and T, and three hydrogen bonds between G and C.
- Due to complementary base pairing the amount of purines and pyrimidines in DNA are equal. The ratio between Adenine and Thymine, Guanine and Cytosine are constant and equal to one (i.e., amount of  $A=T$ , and  $G=C$ ) this is called **Chargaff's rule of base equivalence**.
- The two chains are coiled in a right handed fashion
- The pitch of the helix is 3.4 nm or 34 Å & there are 10 bp in each turn,
- Double stranded DNA molecule has a diameter of 20 Å and distance between two successive base pairs is 0.34 nm (3.4 Å)
- The plane of one base pair stacks over the other in a double helix. This provides stability to the helix along with hydrogen bonding.

#### NOTE -

- Francis Crick proposed the **Central dogma in molecular biology**, which states that the genetic information flows from  $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{Protein}$ .
- In 1970 **Temin** and **Baltimore** discovered that the genetic information also flows in reverse direction i.e., DNA is synthesized from RNA by a process called **reverse transcription** also known as **Teminism**.



#### Packaging of DNA Helix

- Distance between two consecutive base pairs in a DNA =  $0.34 \text{ nm} = 0.34 \times 10^{-9} \text{ m}$
- Total number of base pairs in a human DNA =  $6.6 \times 10^9 \text{ bp}$
- Total length of human DNA = Total no. of base pair  $\times$  Distance between two consecutive base pairs.  

$$= 6.6 \times 10^9 \text{ m} \times 0.34 \times 10^{-9} \text{ m}$$

$$= \sim 2.2 \text{ m}$$
- 2.2 m is too large to be accommodated in the nucleus ( $10^{-6} \text{ m}$ ).

**NOTE –**

- The length of DNA is defined as number of nucleotides (or a pair of nucleotide referred to as base pairs)
- Length is also the characteristic feature of organism
- Example – Bacteriophage  $\Phi$  X 174 has 5386 nucleotides, Bacteriophage lambda has 48502 bp, *Escherichia coli* has  $4.6 \times 10^6$  bp and haploid content of human DNA is  $3.3 \times 10^9$  bp.

The length of *E. coli* DNA is 1.36 mm = Total no. of base pairs x Distance between two consecutive base pair

$$1.36 \text{ mm} = \text{Total no. of base pairs} \times 0.34 \times 10^{-9} \text{ m}$$

$$\text{Total no. of base pairs} = \frac{1.36 \times 10^{-3} \text{ m}}{0.34 \times 10^{-9} \text{ m}}$$

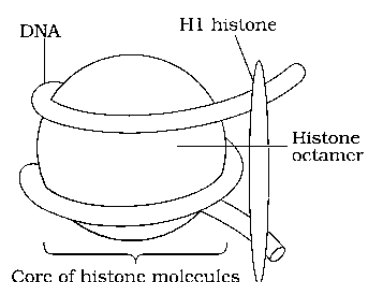
$$\text{Total no. of base pairs} = 4 \times 10^6$$

- **Organisation of DNA in prokaryotes:**

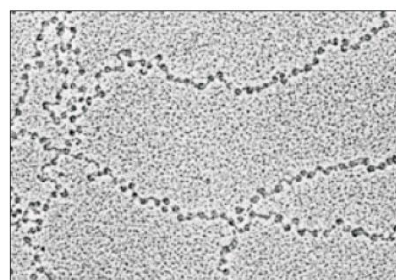
- They do not have nucleus. DNA is scattered.
- In certain regions called nucleoids, DNA (negatively charged) is organized in large loops and is held by some proteins (positively charged).

- **Organisation of DNA in eukaryotes:**

- They have positively charged basic proteins called **histones** (positive and basic due to presence of positive and basic amino acid residues, lysine and arginine).
- **Histone octamer** – Unit of eight molecules of histone
- DNA (negatively charged) winds around histone octamer (positively charged) to form **nucleosome**.
- 1 nucleosome has approx. 200 bp of DNA.
- Nucleosomes constitute the repeating unit of a structure in nucleus called **chromatin** which resemble like **beads present on strings**.

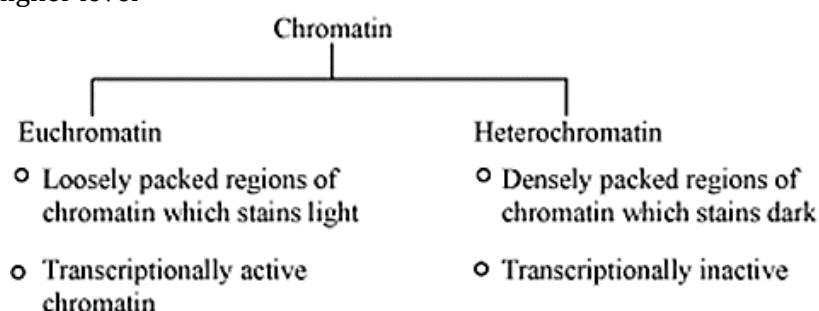


**Figure: Nucleosome**



**Figure: Electron microscope (EM) picture - 'Beads-on-string'**

- Beads on string structure in chromatin are further packaged to form chromatin fibres, which further coil and condense to form chromosomes during metaphase.
- **Non-histone chromosomal (NHC) proteins** – Additional set of proteins required for packaging of chromatin at higher level



## Discovery of DNA as a Genetic Material

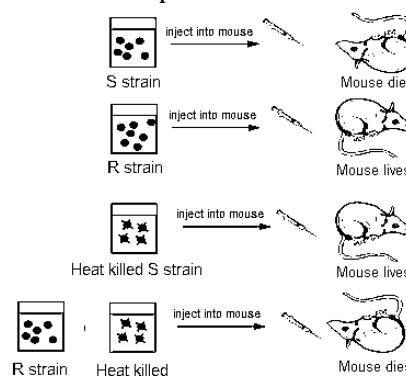
- Though principles of inheritance and discovery of chromosomes in nucleus were achieved long time back, there was confusion about which molecule acted as genetic material.

## Transforming Principle

- In 1928, **Frederick Griffith** performed experiments with the bacteria *Streptococcus pneumoniae*. This bacterium has two strains – S strain and R strain.

| S strain Bacteria                          | R strain Bacteria                         |
|--------------------------------------------|-------------------------------------------|
| • Produce smooth colonies on culture plate | • Produce rough colonies on culture plate |
| • Have a polysaccharide coat               | • Do not have a polysaccharide coat       |
| • Virulent (causes pneumonia)              | • Non-virulent (does not cause pneumonia) |

- Griffith showed that injections of S – strains (virulent) into mice died due to pneumonia disease.
- An injection of R – strains does not produce the pneumonia.
- The heat killed S – strains does not produce the pneumonia.
- The mixture of live R – strains and heat killed S-strains injected into mice died due to pneumonia.
- Griffith recovered living S – strains of bacteria from the dead mice and concluded the some transforming principle transferred from the heat killed S – strains to produce mucous coat in R – strain and they become virulent. The transforming principle is genetic material, but he fail to define biochemical nature of genetic material through his experiments.
- However, the biochemical nature of genetic material was not defined from his experiments.



## Biochemical Characterization of Transforming Material

- Avery, McLeod, and McCarty** (1933 - 44) worked to determine the biochemical nature of genetic material responsible for transformation.
- They purified biochemical (proteins, DNA, RNA etc) from heat killed S cells to see which ones could transform live R cells into S cells.
- They discovered that DNA alone from S bacteria caused R bacteria to become transformed.
- They also discovered that protein-digesting enzymes (proteases) and RNA- digesting enzymes (RNases) did not affect transformation. So the transforming substance was not a protein or RNA.
- Digestion with DNase did inhibit transformation, suggesting that the DNA caused the transformation.
- They concluded that DNA is the hereditary material.

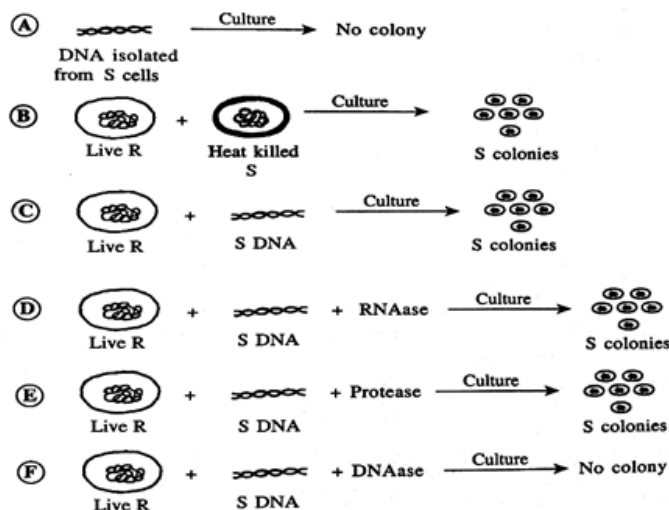


Figure: A generalised presentation of Avery, Macleod and McCarty experiments to show the role of DNA as the genetic material.

## Hershey and Chase Experiment to Confirm DNA as the Genetic Material

- In 1952 **Hershey** and **Chase** worked on bacteriophages (viruses that infect bacteria).
- When a bacteriophage infects a bacterium, the viral genetic material gets attached with the bacterial genetic material and bacteria then treats the viral genetic material as its own to synthesise more viral particles.
- Hershey and Chase worked to discover whether it was a protein or DNA that entered the bacteria from virus.
- They labelled some phages with radioactive sulphur ( $^{35}\text{S}$ ) and the others with radioactive phosphorus ( $^{32}\text{P}$ ).
- These radioactive phages were used to infect *E. coli*.
- *E. coli* was then blended and centrifuged to remove viral particles.
- It was observed that bacteria with radioactive DNA ( $^{32}\text{P}$ ) were radioactive while those with radioactive proteins ( $^{35}\text{S}$ ) lost their radioactivity.
- This showed that it is the DNA that enters the bacteria from viruses and not proteins. Hence, it was concluded that DNA is the genetic material.

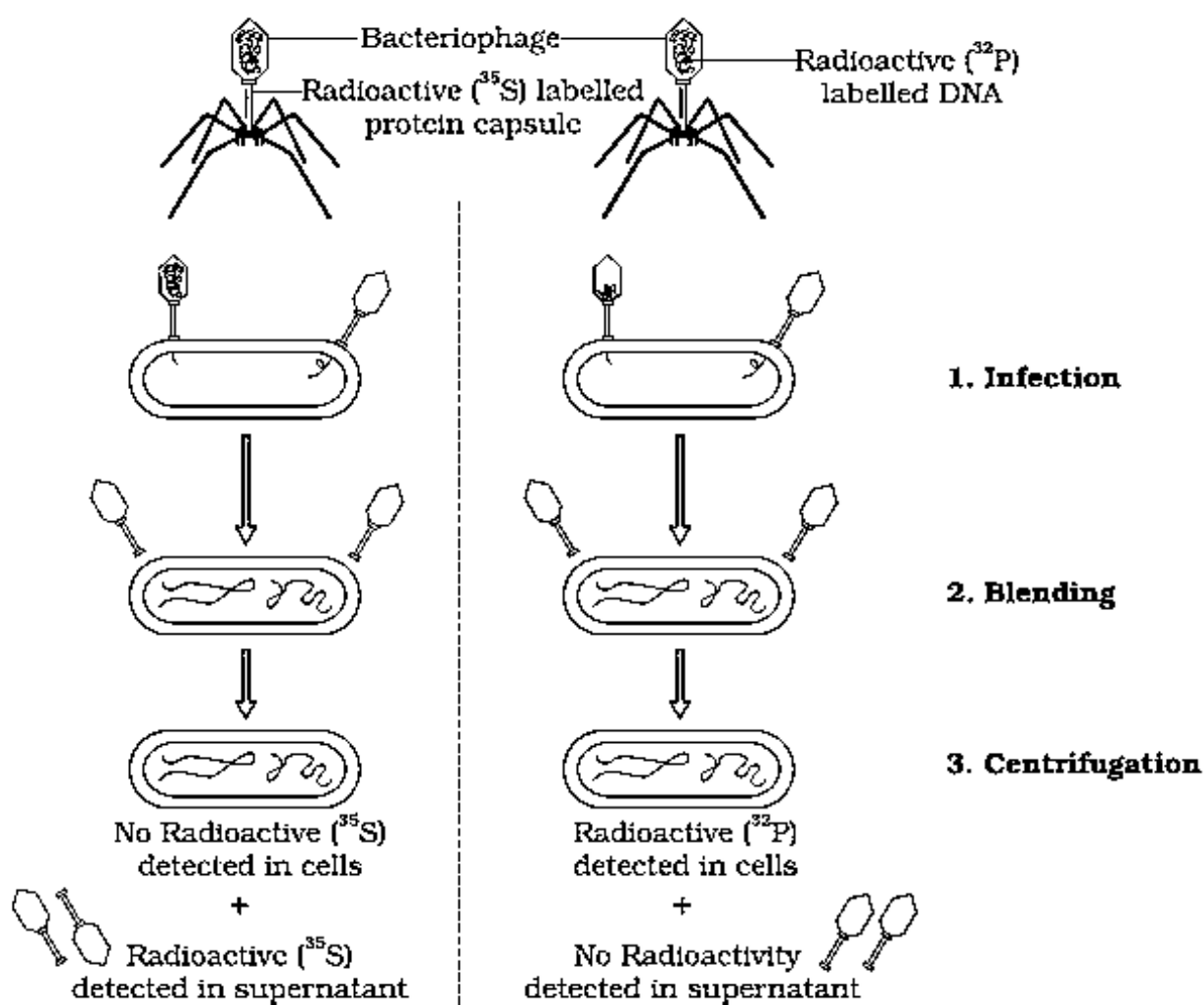


Figure: The Hershey-Chase experiment

## Properties of the Genetic Material

- It should be able to replicate (duplicate to produce its identical copy).
- It should be chemically and structurally stable.
- It should have scope for changes (mutation) that are essential for evolution.
- It should follow the Mendelian principles of inheritance.

**Difference between DNA and RNA:**

| DNA                                                                                      | RNA                                                                                                                                                             |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| • Has deoxyribose sugar                                                                  | • Has ribose sugar                                                                                                                                              |
| • 5-methyl uracil (thymine) is present.                                                  | • Uracil is present in place of thymine.                                                                                                                        |
| • Mostly DNA acts as the genetic material.                                               | • RNA acts as a messenger and adaptor (tRNA). It acts as a genetic material in some viruses ( <b>Tobacco Mosaic Virus, Q<math>\beta</math> Bacteriophage</b> ). |
| • DNA is stable.                                                                         | • Presence of 2' OH group at every nucleotide makes RNA labile and easily biodegradable.                                                                        |
| • Chemically less reactive, mutates slowly                                               | • Mutation in RNA is faster.                                                                                                                                    |
| • DNA requires RNA for protein synthesis.<br>DNA $\rightarrow$ RNA $\rightarrow$ Protein | • RNA directly codes for proteins.                                                                                                                              |

**Why DNA is more stable than RNA?**

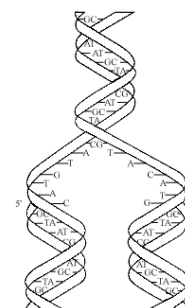
- In RNA, a 2' OH group is present at every nucleotide. This makes RNA unstable and degradable.
- Presence of thymine in place of uracil confers additional stability to DNA.
- RNA being a biocatalyst is more reactive.
- DNA is double-stranded having complementary strand, which further resists the changes by repair mechanism.

**NOTE –**

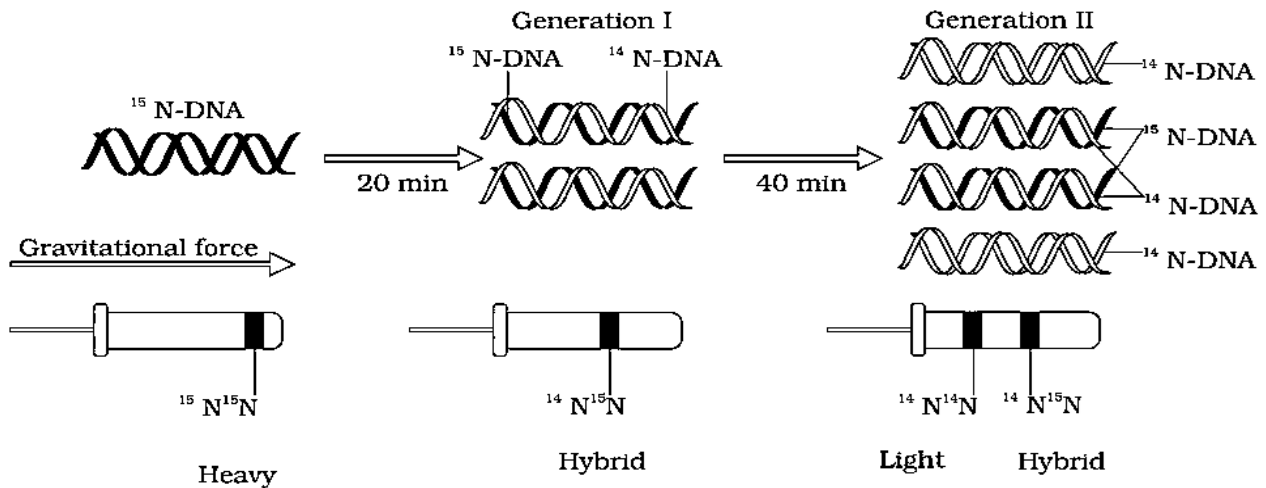
- DNA being more stable is preferred for storage of genetic information.
- For the transmission of genetic information RNA is better.
- **RNA was the first genetic material** because metabolism, translation, splicing etc. evolved around RNA.
- DNA evolved from RNA by certain modification.

**What is DNA Replication? (The description below is of Semi conservative mode)**

- DNA replication is the phenomenon in which a duplicate copy of DNA is synthesized.
- In replication, two strands of the DNA helix separate and each strand acts as a template for synthesising new complementary strands.
- After completion of replication, the two copies so produced will have one parental and one newly synthesised strand. This scheme of replication is called **semi-conservative replication**.
- Semi-conservative replication was **proposed by Watson and crick** in 1953 immediately proposing the double helical structure for DNA

**Experiment to Prove That DNA Replicates Semi-Conservatively**

- Performed by – **Messelson and Stahl** in 1958
- *E.coli* was grown in a medium containing heavy isotope  $^{15}\text{N}$  ( $^{15}\text{NH}_4\text{Cl}$ ) as the nitrogen source.
- $^{15}\text{N}$  was incorporated into newly synthesised DNA as well and the DNA became heavy DNA.
- Heavy DNA molecule can be differentiated from normal DNA by density gradient centrifugation using **cesium chloride (CsCl)** as the gradient. (**Note** –  $^{15}\text{N}$  is not a radioactive isotope, and it can be separated from  $^{14}\text{N}$  only based on densities).
- Then, cells were again transferred into a medium with  $^{14}\text{N}$  as nitrogen source. Samples were taken from this media and their DNA was extracted.
- *E.coli* divides every 20 minutes. Therefore, the DNA extracted after 20 minutes had a hybrid density.
- DNA extracted after 40 minutes had equal amount of hybrid and light intensities.  
This implies that the newly synthesised DNA obtained one of its strands from the parent. Thus, replication is semi-conservative.



(Separation of DNA by Centrifugation)

**NOTE -**

- Very similar experiments involving use of radioactive thymidine to detect distribution of newly synthesised DNA in the chromosomes was performed on *Vicia faba* (faba beans) by Taylor and colleagues in 1958.
- The experiments proved that the DNA in chromosomes also replicate semi-conservatively.

**Mechanism of DNA Replication**

- Replication occurs in S phase of cell cycle.
- Enzyme involved - **DNA polymerase** (DNA dependent DNA polymerase), **Helicase**, **Topoisomerase**, **RNA primase**, **Ligase**.
- Replication requires energy.
- Source of energy – **Deoxyribonucleoside triphosphates (DNTPs)**
- DNTPs have dual purpose – Act as substrates and provide energy also
- Replication initiates at specific regions in DNA called **origin of replication**.
- DNA polymerase polymerises a large number of nucleotides in a very short time.
- During the course of replication, two parent strands do not completely open, but a small opening forms in which replication occurs. This small opening forms a **replication fork**.
- DNA polymerase can polymerise only in one direction that is  $5' \rightarrow 3'$ .
- Therefore, replication occurs smoothly at  $3' \rightarrow 5'$  end of DNA template. (continuous replication, but occurs discontinuously at  $5' \rightarrow 3'$  end)
- The discontinuous fragments (**Okazaki fragments**) so formed are joined by DNA ligase.

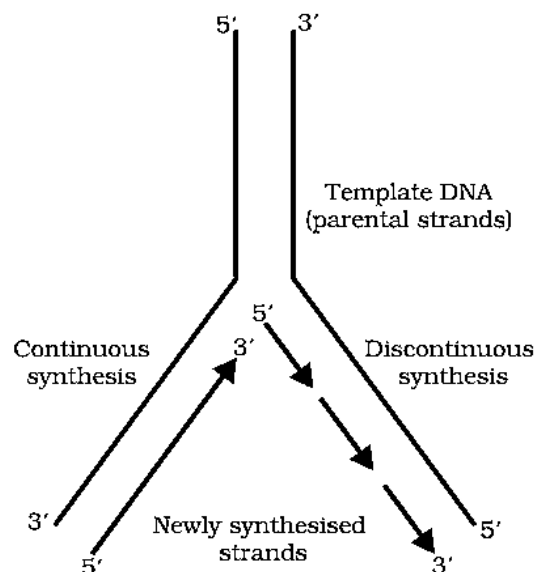


Figure: Replicating Fork

**NOTE -**

- *E.coli* has  $4.6 \times 10^6$  bp (compare it with human whose diploid content is  $6.6 \times 10^9$  bp) and completes the process of replication within 38 min. i.e., Average rate of polymerization has to be approximately 2000 bp/sec

**Transcription**

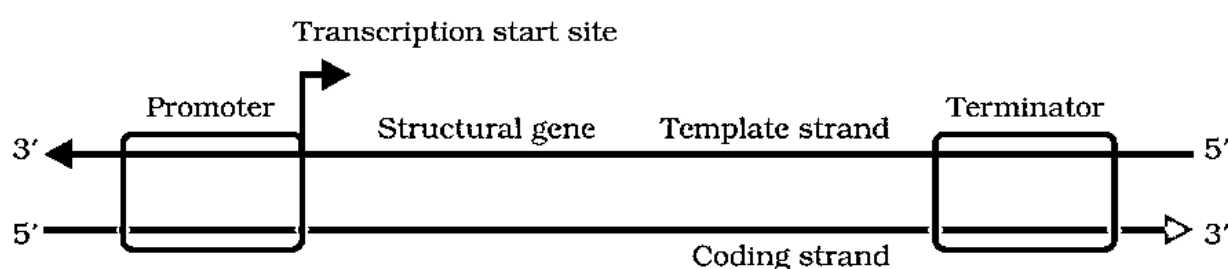
- Transcription is the process of formation of RNA molecules from the DNA.
- During transcription, only a segment of DNA from only one of the strands participates.
- Both strands are not copied during transcription because:
  - If both strands get transcribed at the same time since the sequences of amino acid would be different in both (due to complementarity), then two RNA molecules with different sequences will be formed, which in turn give rise to two different proteins. Therefore, one DNA would end up giving rise to two different proteins.
  - Two RNA molecules so formed will be complementary to each other, hence would end up forming a double-stranded RNA leaving the entire process of transcription futile.

**Transcriptional Unit**

- A transcriptional unit has primarily three regions:
  - Promoter** – Marks the beginning of transcription; RNA polymerase binds here
  - Structural gene** – Part of the DNA that is actually transcribed
  - Terminator** – Marks the end of transcription
- A transcriptional unit is the structural gene flanked by the promoter and terminator

**Template Strand and Coding Strand**

- Enzyme involved in transcription, **RNA polymerase** (DNA dependent RNA polymerase), catalyses in only one direction i.e.,  $5' \rightarrow 3'$ .
- Therefore, the strand with polarity  $3' \rightarrow 5'$  acts as a template (**Template Strand/antisense/noncoding strand**).
- The strand with polarity  $5' \rightarrow 3'$  acts as **coding strand** or **sense strand** (which is a misnomer since it does not code for anything). Coding strand has sequence similar to RNA formed after transcription except for the change that thymine is present instead of uracil.



*Figure: schematic structure of a transcription unit*

**Exercise -**

- A hypothetical sequence from a transcription unit is represented below:

3' -ATGCATGCATGCATGCATGC- 5'    Template Strand

5' -TACGTACGTACGTACGTACG- 3'    Coding Strand

*Can you now write the sequence of RNA transcribed from above DNA?*

**NOTE –**

- The promoter is said to be located towards 5' –end (upstream) of the structural gene (the reference is made with respect to the coding strand)
- The terminator is located towards 3' –end (downstream) of the coding strand
- There are additional regulatory sequences that may be present further upstream or downstream to the promoter.

**Gene**

- A gene is defined as the functional unit of inheritance.
  - The DNA sequence which codes for tRNA or rRNA molecule also defines a gene.
  - **Cistron** – Segment of DNA (structural gene) that contains the genetic code for a polypeptide.
  - The structural genes could be of two types:
    1. **Monocistronic** (mostly in eukaryotes)
    2. **Polycistronic** (mostly in prokaryotes)
1. **Monocistronic genes (split genes)** (mostly in eukaryotes) have two parts:
    - **Exon (coding region)**– Sequences that code for a particular character and is expressed in a matured and processed mRNA
    - **Intron (noncoding region)** – Interrupting sequences that do not appear in a mature and processed mRNA (Introns are transcribed but not translated)
  2. **Polycistronic genes** (mostly in prokaryotes) – have contain more than one structural gene with coding sequence.
- **Regulatory genes** – Sequences that do not code for anything, but have regulatory functions

**Types of RNA**

1. **mRNA** (messenger RNA) – It serves as a template for protein synthesis. DNA is transcribed to form an mRNA, which in turn is translated to form protein. **[Central dogma of molecular biology]**
2. **tRNA** (transfer RNA) – It brings amino acids during translation and reads the genetic code.
3. **rRNA** (ribosomal RNA) – They play a structural and catalytic role during translation.

**Transcription Process**

- Transcription has three steps – **initiation, elongation and termination.**
1. **Initiation:**
    - RNA polymerase binds with the promoter to initiate the process of transcription.
    - Association with initiation factor ( $\sigma$ ) (sigma factor) alters the specificity of RNA polymerase to initiate the transcription.
  2. **Elongation:**
    - RNA polymerase uses nucleotide triphosphate as substrate, and polymerisation occurs according to complementarity.
  3. **Termination:**
    - Termination occurs when termination factor ( $\rho$ ) (Rho factor) alters the specificity of RNA polymerase to terminate the transcription.
    - As the RNA polymerase proceeds to perform elongation, a short stretch of RNA remains bound to the enzyme. As the enzyme reaches the termination region, this nascent RNA falls off and transcription is terminated.

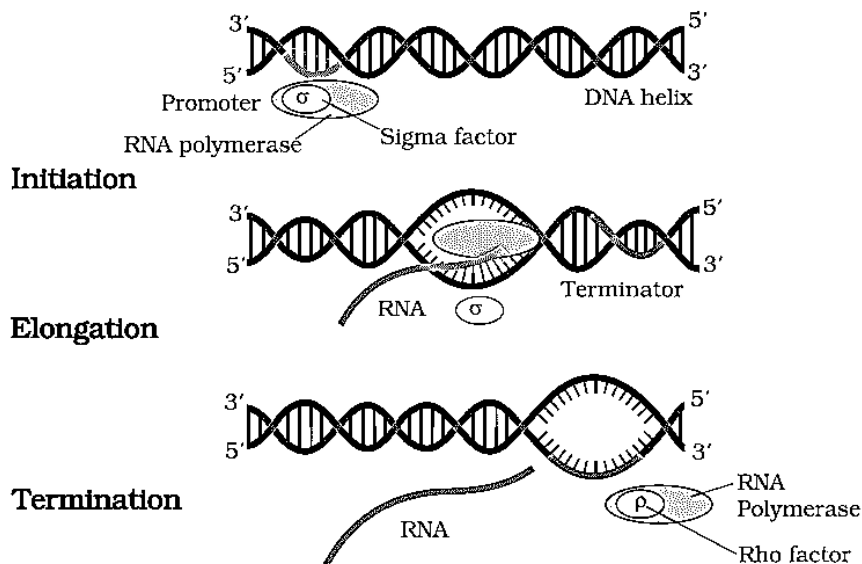


Figure: Transcription process in bacteria

### Complexities Associated with Transcription

#### In prokaryotes:

- There is no clear demarcation between cytosol and nucleus. Therefore, translation can begin even before transcription is completed. Thus, in prokaryotes, transcription and translation are coupled.

#### In eukaryotes:

- Three different kinds of RNA polymerases are present.
  - RNA polymerase I transcribes rRNA (28S, 18S and 5.8S)
  - RNA polymerase II transcribes hnRNA (heterogeneous nuclear RNA) (mRNA precursor).
  - RNA polymerase III transcribes tRNA, 5srRNA, and snRNA (small nuclear RNA).

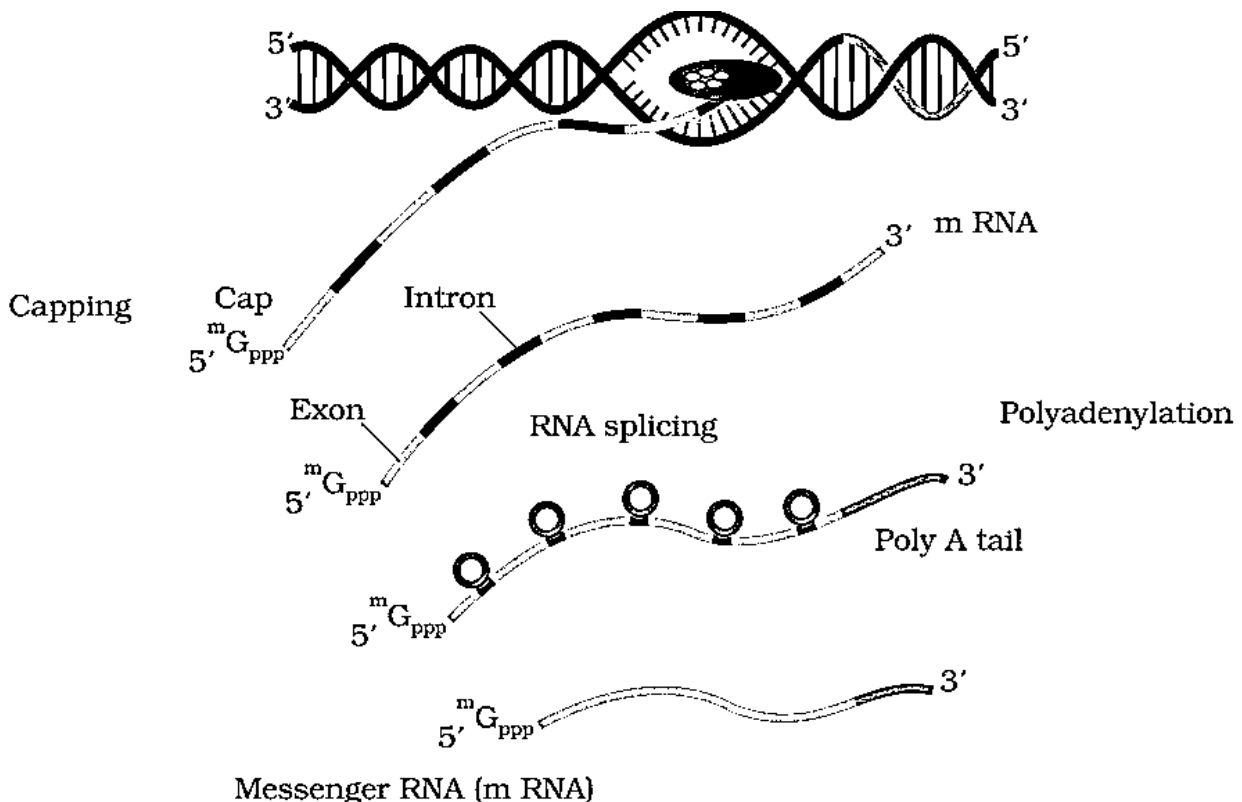


Figure: process of Transcription in Eukaryotes

- The precursor of mRNA, i.e. hnRNA – **primary transcript**, contains both exons and the introns and are non-functional.
- Introns are removed and exons are joined by a process called **splicing**.
- **Capping** – in this, methyl guanosine triphosphate is added to the 5' end of hnRNA.
- **Tailing** – in this, adenylate residues (200 - 300) are added (polyadenylation) to the 3' end of hnRNA.
- When hnRNA is fully processed, it is known as mRNA, which is transported out of the nucleus to get translated.

#### NOTE –

- The split gene arrangements represent probably an ancient feature of genome.
- The presence of introns is reminiscent of antiquity, and the process of splicing represents the dominance of **RNA-world**.

#### Genetic Code

- Genetic code directs the sequence of amino acids during the synthesis of proteins.
- **George Gamow proposed** that if 20 amino acids are to be coded by 4 bases, then the code should be made up of three nucleotides.  $4^3 = 64$  ( $4^1 = 4$  or  $4^2 = 16$ , which is less than 20) so, the codon was proposed to be triplet.
- **Har Gobind Khorana** developed a chemical method to synthesise RNA molecules with defined combination of bases.
- **Nirenberg** developed cell-free systems for protein synthesis, which helped the code to be deciphered.
- The enzyme known as **Severo Ochoa enzyme** (polynucleotide phosphorylase) helped to polymerise RNA with defined sequences in a template independent manner.
- It finally gave rise to the checker-board for genetic code.

#### The codons for the Various Amino Acids

| First position | Second position |         |          |          | Third position |
|----------------|-----------------|---------|----------|----------|----------------|
|                | U               | C       | A        | G        |                |
| U              | UUU Phe         | UCU Ser | UAU Tyr  | UGU Cys  | U              |
|                | UUC Phe         | UCC Ser | UAC Tyr  | UGC Cys  | C              |
|                | UUA Leu         | UCA Ser | UAA Stop | UGA Stop | A              |
|                | UUG Leu         | UCG Ser | UAG Stop | UGG Trp  | G              |
| C              | CUU Leu         | CCU Pro | CAU His  | CGU Arg  | U              |
|                | CUC Leu         | CCC Pro | CAC His  | CGC Arg  | C              |
|                | CUA Leu         | CCA Pro | CAA Gin  | CGA Arg  | A              |
|                | CUG Leu         | CCG Pro | CAG Gin  | CGG Arg  | G              |
| A              | AUU Ile         | ACU Thr | AAU Asn  | AGU Ser  | U              |
|                | AUC Ile         | ACC Thr | AAC Asn  | AGC Ser  | C              |
|                | AUA Ile         | ACA Thr | AAA Lys  | AGA Arg  | A              |
|                | AUG Met         | ACG Thr | AAG Lys  | AGG Arg  | G              |
| G              | GUU Val         | GCU Ala | GAU Asp  | GGU Gly  | U              |
|                | GUC Val         | GCC Ala | GAC Asp  | GGC Gly  | C              |
|                | GUA Val         | GCA Ala | GAA Glu  | GGA Gly  | A              |
|                | GUG Val         | GCG Ala | GAG Glu  | GGG Gly  | G              |

**Salient features of genetic code:**

- **Codon is triplet** – three adjacent nitrogen bases constitute a codon which specifies the placement of one amino acid in a polypeptide.
- **Codons are unambiguous** - One codon codes for a single specific amino acid.
- **Codons are degenerate** - Some amino acids are coded by more than one codon.
- **Genetic code is universal** - 1 codon codes for same amino acid in all species.
- **Codons are commaless** - Codons are read continuous in mRNA. They lack punctuations.
- **AUG has dual functions** – Acts as a **start** or **initiator codon** and codes for Methionine (Met).

**NOTE –**

- 61 codons code for amino acids while 3 are stop codons
- **Start codon** – AUG and sometimes GUG
- **Stop codon** – UAA (ochre), UAG (amber), UGA (opal) – they do not specify any amino acid and are hence also called **nonsense codons** or **terminator codon**.

**Extra information - Exception of Genetic Code**

- **Different codons** – In *paramecium* and some other ciliates termination codons UAA and UGA code for glutamine.
- **Mitochondrial genes** – AGG and AGA code for arginine but function as stop signals in human mitochondrion. UGA, a termination codon, corresponds to tryptophan while AUA (codon for isoleucine) denotes methionine in human mitochondria.

**Exercise –**

If the following is the sequence of nucleotides in mRNA, predict the sequence of amino acid coded by it (take help of the checkerboard)

*-AUG UUU UUC UUC UUU UUU UUC-*

Now try the opposite. Following is the sequence of amino acids coded by mRNA. Predict the sequence of nucleotide in the RNA:

*Met-Phe-Phe-Phe-Phe-Phe-Phe*

*Do you face any difficulty in predicting the opposite?*

*Can you now correlate which two properties of genetic code you have learnt?*

**Effects of Mutations on Genetic Code**

- Mutations include insertions, deletions, and rearrangements.
- Mutation results in changed phenotype and diseases such as sickle cell anaemia. (Change Glu → Val in gene (CTC → CAC) coding for beta globin chain of haemoglobin) Such mutations are called **point mutations**.
- Insertion or deletion of a single base pair disturbs the entire reading frame in mRNA. Such mutations are called **Frameshift mutations**.
- Frameshift mutations hold the proof of the fact that codon is triplet because if we insert three or multiple of three bases followed by the deletion of same number of bases, then the reading frame will remain unaltered.

**Exercise –**

Consider a statement that is made up of the following words each having three letters like genetic code.

**RAM HAS RED CAP**

If we insert a letter B in between HAS and RED and rearrange the statement, it would read as follows:

**RAM HAS BRE DCA P**

Similarly. If we now insert two letters at the same place, say BI'. Now it would read,

**RAM HAS BIR EDC AP**

Now we insert three letters together, say BIG, the statement would read

**RAM HAS BIG RED CAP**

The same exercise can be repeated, by deleting the letters R, E and D, one by one and rearranging the statement to make a triplet word.

**RAM HAS EDC AP**

**RAM HAS DCA P**

**RAM HAS CAP**

**Structure of tRNA**

- tRNA is an adapter molecule. On one hand, it reads the genetic code and on the other hand, it binds to specific amino acids.
- tRNA has an **anticodon loop** that has bases complementary to the mRNA code and an **amino acid acceptor end (3' end)** where it binds to the corresponding amino acid.
- Initiation tRNA** – this tRNA is essential for initiation of translation and has UAC in anticodon loop and Met in amino acid acceptor end.
- There are no tRNAs for stop codons.

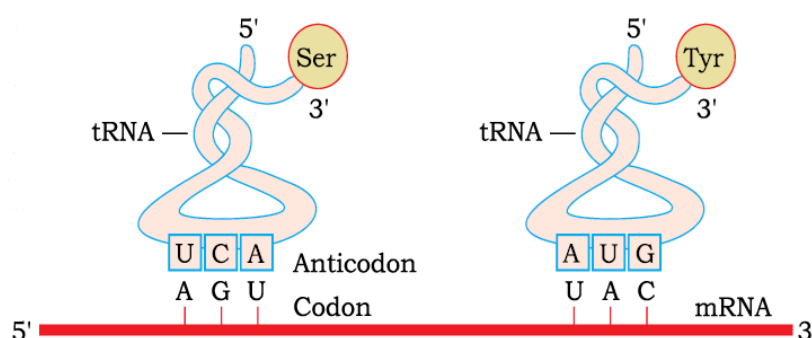
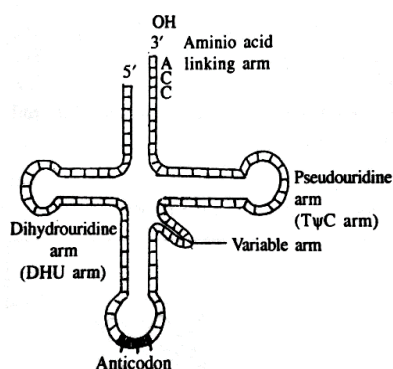


Figure: tRNA - the adapter molecule

**Translation**

- It is the process where coded information present in the triplets (codons) of mRNA is decoded into specific protein molecule
- The translation process takes by following steps.

**Activation of amino acids** - amino acids are activated in the presence of ATP.

**Charging of tRNA or amino-acylation of tRNA –**

- Activated amino acids are linked to their corresponding tRNA (amino acyl tRNA).
- Ribosomes are the site for translation (protein synthesis). Ribosomes (consist of rRNA and 80 different protein) have 2 subunits: a large subunit and a small subunit.

### Initiation of polypeptide chain

- The smaller subunit binds to the mRNA at the start codon (AUG) that is recognized only by the initiator tRNA forming **initiation complex**.
- After formation of initiation complex large subunit of ribosome units with small subunit

### Elongation of polypeptide chain

- The second t-RNA-amino acid complexes bind to the second codon on the mRNA and base pairing occurs between codon on mRNA and tRNA anticodon.
- The peptide bond is formed between first amino acid and second amino acid (The ribosome act as a catalyst (23S rRNA in bacteria is the enzyme – ribozyme) for the formation of peptide bond)
- The ribosomes moves from codon to codon on the mRNA in the direction  $5' \rightarrow 3'$  and amino acids are added one by one.

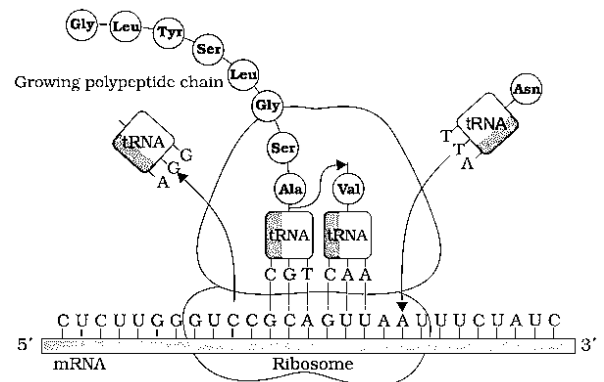


Figure: Translation

### Termination

- Release factor** binds to stop codon to terminate the translation.

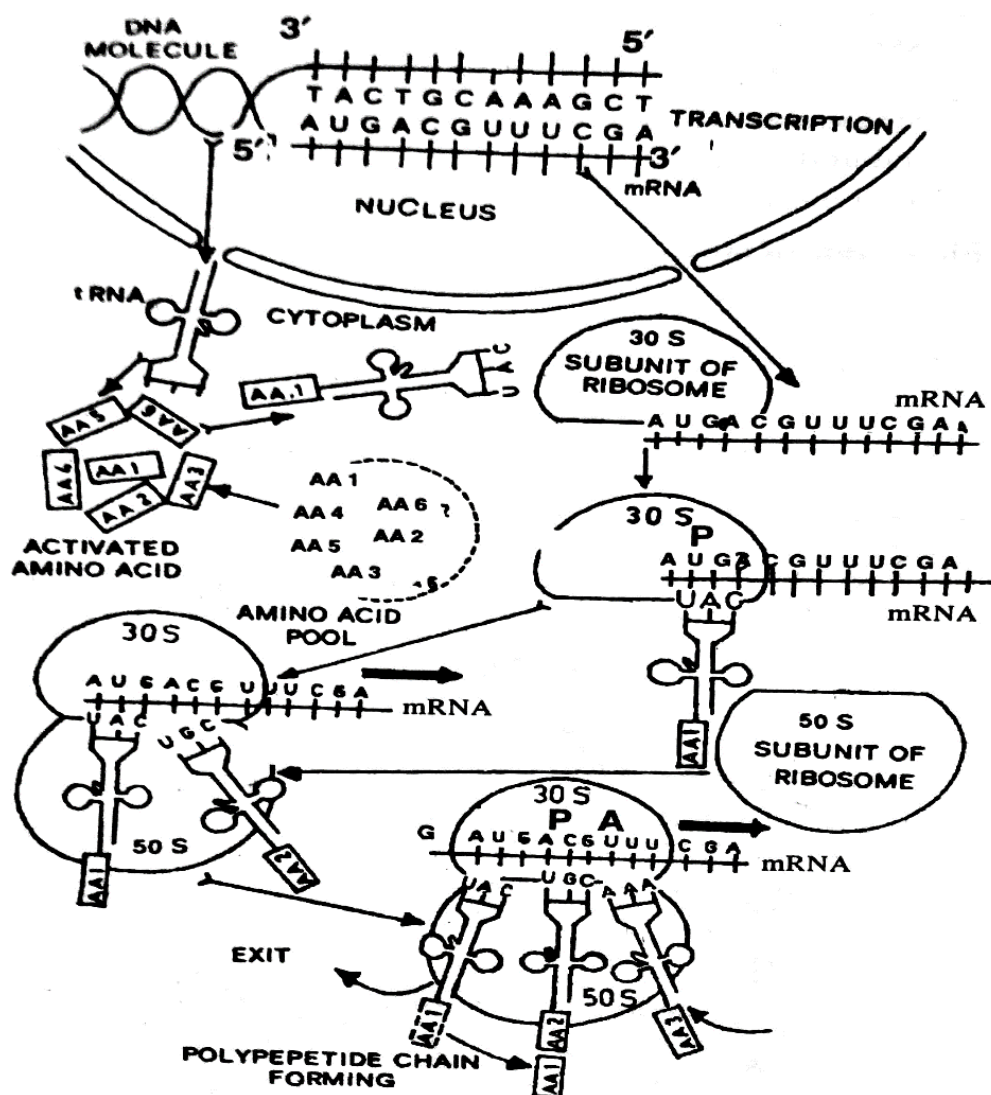
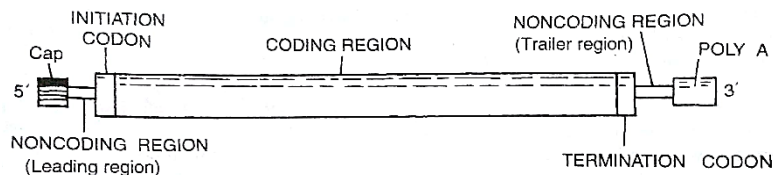


Figure: diagrammatic representation of overall summary of protein synthesis in *E.coli*

**NOTE –**

- Translational unit in an mRNA is the region flanked by start codon and stop codon.
- **Untranslated regions** (UTR) are the regions on mRNA that are not themselves translated, but are required for efficient translation process. They may be present before start codon (5' UTR) or after stop codon (3' UTR).

**Regulation of Gene Expression**

- Regulation of gene expression could be exerted at following levels.
  - Transcriptional level (following of primary transcripts)
  - Processing level (splicing)
  - Transport of mRNA from nucleus to cytoplasm
  - Translational level
- In addition, metabolic, physiological, or environmental conditions regulate the expression of genes.

**Regulation in prokaryotes**

- **Operon** – An arrangement where a polycistronic gene is regulated by a common promoter and regulatory genes
- Example - *Lac* operon, *trp* operon, *his* operon, *val* operon.
- Operons are of two types – **inducible** (occur in catabolic pathway) and **repressible** (occur in anabolic pathway).
- Gene expression is regulated by controlling the rate of transcriptional initiation.
- The activity of RNA polymerase at a given promoter is regulated by accessory proteins.
- A regulatory protein could be activator or repressor.
- Operator is the region located adjacent to promoter.
- Each operon has a specific operator and a specific repressor.
- Usually a repressor protein binds to operator gene.

**Regulation of *Lac* Operon – Inducible Operon System.**

- The elucidation of *lac* operon as a transcriptionally active system was first done by geneticist **Jacob** and biochemist **Monod** in 1961.
- An inducible operon system is a regulated unit of genetic material which is switched on in response to the presence of chemical (inducer).
- Genes constituting *lac* operon:

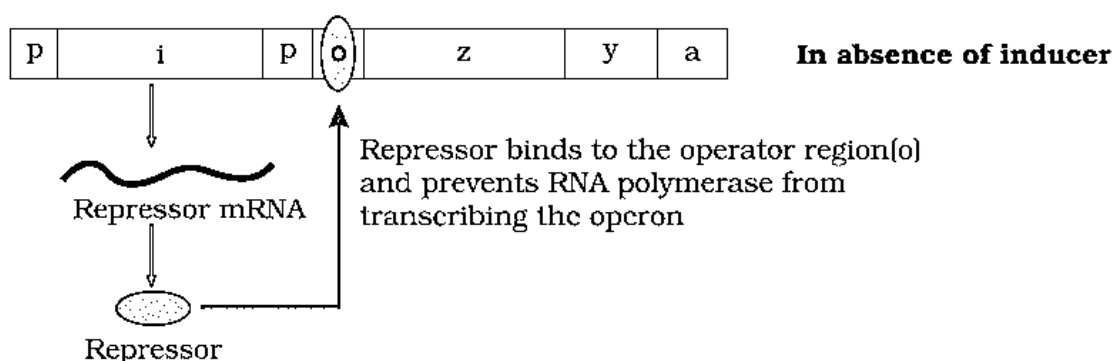
| Gene          | Nature     | Function                                                                             |
|---------------|------------|--------------------------------------------------------------------------------------|
| <i>i</i> gene | Inhibitor  | It codes for repressor of <i>lac</i> operon.                                         |
| <i>P</i> gene | Promoter   | Binding site for RNA polymerase enzyme.                                              |
| <i>O</i> gene | Operator   | Regulates the transcription of structural genes.                                     |
| <i>z</i> gene | Structural | It codes for $\beta$ -galactosidase. Which digest lactose into glucose and galactose |
| <i>y</i> gene | Structural | It codes for permease, which increases the permeability of cell to lactose.          |
| <i>a</i> gene | Structural | It codes for transacetylase (function is unknown)                                    |

- All genes involved in *lac* operon are required for metabolism of lactose.
- Expression of genes coding for enzymes is required only when substrate for that enzyme is available.  
For example –  $\text{Lactose} \xrightarrow{\beta\text{-galactosidase}} \text{Glucose} + \text{Galactose}$

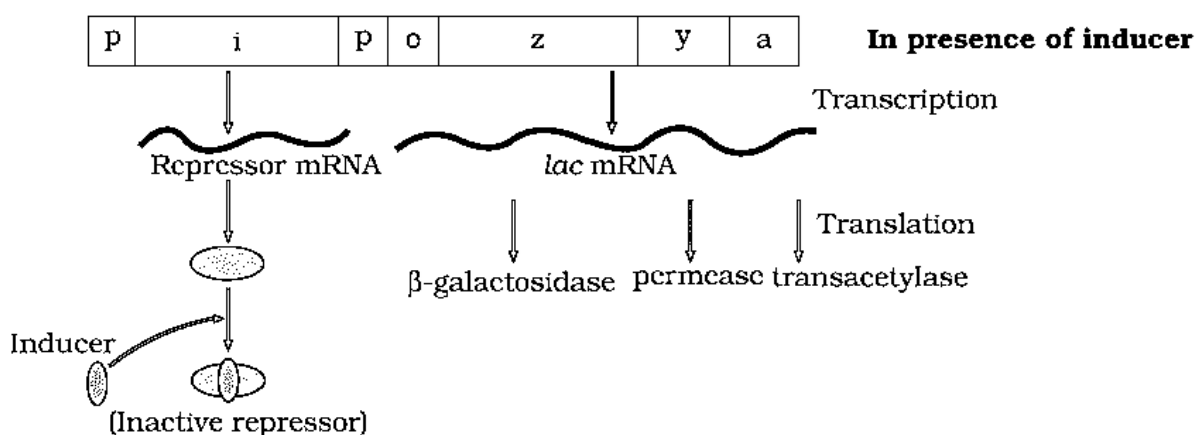
- E.coli synthesizes  $\beta$ -galactosidase, only when lactose is available.
- **Inducer** – Lactose (also allolactose or metabolite of lactose) acts as an inducer for *lac* operon since it regulates the switching on and off of the operon.
- If lactose is provided to the growth media of bacteria in absence of any other carbon source, then it is transported inside the cells by permease.
- For permease to be present and lactose to enter inside the cells, low level of expression of *lac* operon must be present all the time.

### Regulation of *Lac* Operon in Absence of Inducer – Negative Regulation

- In absence of inducer (lactose), *i* gene transcribes to synthesize repressor mRNA, which translates to form repressor protein.
- This repressor protein binds with the operator region of operon and prevents RNA polymerase to transcribe genes – *z*, *y*, and *a*. Hence, regulation of *lac* operon by repressor is referred to as **negative regulation**.
- Therefore, in absence of the products of these genes, metabolism of lactose ceases.



### Regulation of *Lac* Operon in Presence of Inducer.



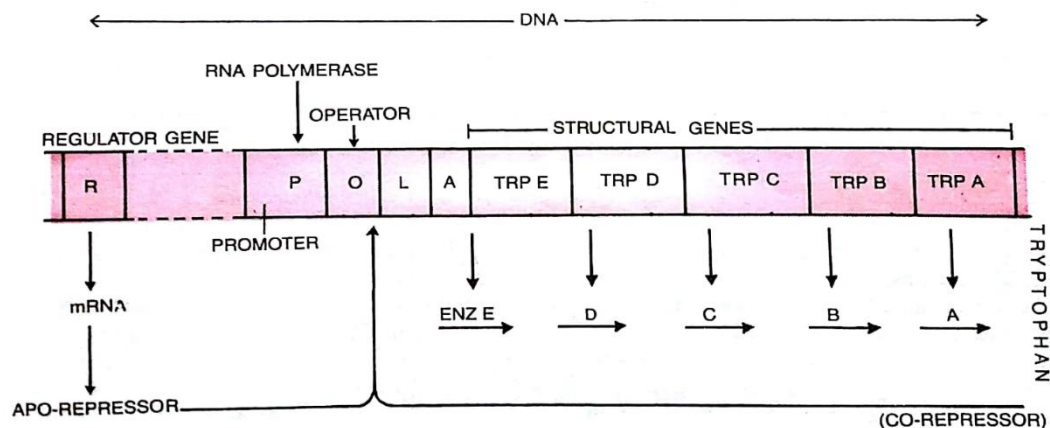
- In presence of inducers such as lactose or allolactose, the repressor is inactivated by interaction with the inducer.
- This allows RNA polymerase access to the promoter and transcription proceeds.
- In presence of all these enzymes, the metabolism of lactose proceeds in a normal manner.

### Extra Information

#### Repressible operon system – Trptophan or *trp* operon

- Repressible operon system is commonly found in **anabolic pathways**.
- The operon is active and the enzymes produced by its structural genes are normally present in the cell.

- The functioning of the operon is stopped when the concentration of an end product crosses a threshold value. An example of repressible system is **tryptophan** or **trp operon** of *Escherichia coli*.
- It was worked out by Jacob and Monod and consists of the following:



- **Structural Genes -**
  - The genes are connected to transcription of mRNAs. The mRNAs translate their coded information in the synthesis of polypeptides.
  - Polypeptides give rise to proteinaceous substances including enzymes.
  - Tryptophan operon has five structural genes - **trp, E, D, C, B, A**. They form enzymes for five steps of tryptophan synthesis.
- **Operator Genes -**
  - It controls the functioning of structural genes.
  - Normally it is kept switched on because the aporepressor produced by regulator gene is unable to completely block operator gene.
  - The operator gene is switched off when a corepressor is available along with aporepressor.
- **Other Regulatory Region Components -**
  - Two components of regulatory region occur between operator gene and structural gene E.
  - They are leader sequence (L) and attenuator (A). The leader sequence is involved in controlling attenuator.
  - Attenuator helps in reducing tryptophan synthesis when it is present in sufficient amount without switching off the operon.
- **Regulatory Gene (*trp R*) -**
  - It forms a proteinaceous component for possible blocking the activity of operator gene.
  - Regulator gene of tryptophan operon lies at a distance away from the remaining operon.
- **Aporepressor -**
  - It is a proteinaceous substance synthesised by regulator gene.
  - Aporepressor forms a component of repressor for blocking the working of operator gene. For this it requires a corepressor.
  - When corepressor is not available in proper strength, the operator gene is kept switched on because by itself, aporepressor is unable to block the working of operator gene.

- **Corepressor –**

- It is a nonproteinaceous component of repressor which is also an end product of reactions catalysed by enzyme produced through the activity of structural genes.
- The end product is often utilized in some other reaction so that it rarely accumulates and hence does not function as corepressor.
- However, whenever it accumulates or becomes available from outside source, the end product becomes corepressor, combines with aporepressor, forms repressor and blocks the operator gene.
- The structural gene now stop transcription.
- The phenomenon is known as **feed-back repression**. It exerts a **negative control**.
- In tryptophan operon, tryptophan (an amino acid) functions as corepressor.

### Human Genome Project (HGP)

- It was a 13-year project coordinated by US department of energy and National Institute of Health (NIH); later joined by Wellcome Trust (UK)
- Launched in 1990, completed in 2003

### Goals of HGP

- Identify all the approximately 20,000 – 25,000 genes in human DNA.
- Determine the sequences of the 3 billion chemical base pairs that make up human DNA.
- Store this information in database.
- Transfer related technologies to other sectors, such as industries.
- Address the ethical, legal and social issues (**ELSI**) that may arise from the project.
- Determining its sequence of base pairs can solve many medical, agricultural, environmental, and evolutionary mysteries.

### Relationship of HGP with Bioinformatics

- Human genome (genome refers to the totality of genes that are present in a human being) contains  $3 \times 10^9$  base pairs.
- Cost of sequencing 1 bp = US \$ 3
- Cost of sequencing  $3 \times 10^9$  bp = US \$ 9 billion
- Enormous sequence data so generated would have required 3300 books containing 1000 pages each just for a human genome.
- Hence, a new branch of biology has been developed known as **bioinformatics** for storing, retrieving, and analysing enormous data
- Genomes of many non-human models such as bacteria, yeast, *Caenorhabditis elegans* (a free living non-pathogenic nematode), *Drosophila* (fruit fly), plants (rice and *Arabidopsis*) have also been sequenced.

#### EXTRA INFORMATION

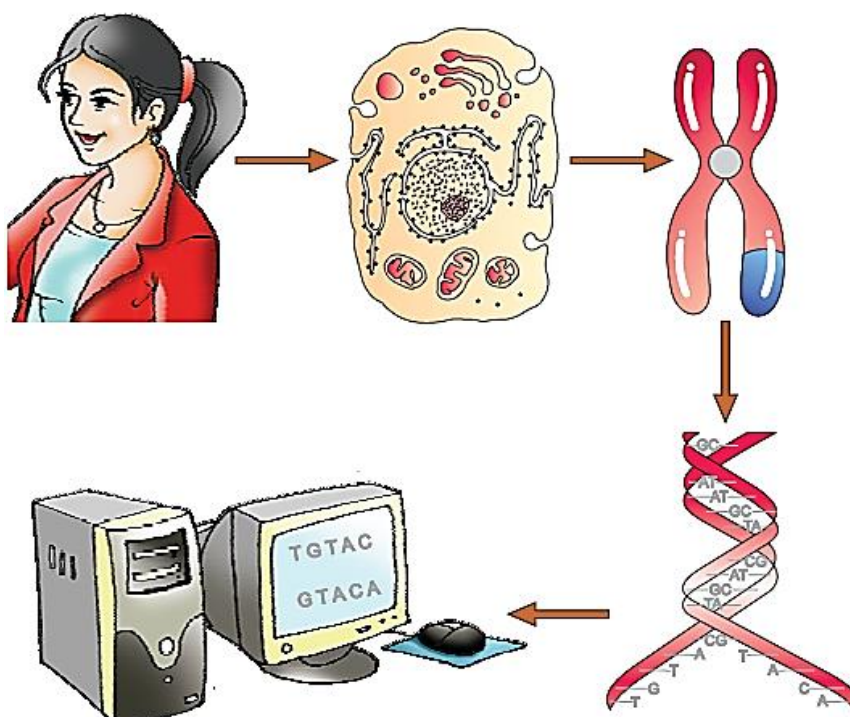
- The whole genome to be sequenced was the bacterium – *Haemophilus influenza*.
- The round worm *Caenorhabditis elegans* was the first animal to have its whole genome sequenced.
- In plants – *Arabidopsis* was the first to have its whole genome sequenced.
- *Drosophila melanogaster*'s whole genome was sequenced in the year 2000.
- The sequence of chromosome 1 of human was completed in May 2006 (this was the last of the 24 human chromosomes – 22 autosomes and X and Y – to be sequenced).

## Methods to Identify Genes

- Two methods – identifying ESTs (Expressed sequence Tags) and sequence annotation.
- **ESTs (Expressed Sequence Tags)** – as the name suggests, this refers to the part of DNA that is expressed, i.e. transcribed, as mRNA and translated into proteins thereafter. It basically focuses on sequencing the part denoting a gene.
- **Sequence Annotation** – in this approach, entire genome (coding + non-coding) is sequenced and later on function is assigned to each region in the genome.

## Genome Sequencing

- DNA from the cells is isolated and is randomly broken into fragments of smaller sizes.
- These fragments are cloned into suitable host using vectors (it is DNA molecule used as a vehicle to artificially carry foreign genetic material into another cell where it can be replicated).
- Cloned fragments amplify in the host. Amplification facilitates an easy sequencing.
- Common vectors used – **BAC** (Bacterial artificial chromosomes) and **YAC** (Yeast artificial chromosomes)
- Common hosts – Bacteria and yeasts
- Automated sequencers are used to sequence these smaller fragments (**Sanger sequencing**).
- The sequences so obtained are arranged based on overlapping regions within them (alignment).
- Alignment of the sequences is also done automatically by computer programs.
- Then these sequences are annotated and assigned to each chromosome.



*Figure: A representative diagram of human genome project*

## Preparation of Genetic and physical maps on Genome

- 2 methods are used – restriction polymorphism and microsatellites
- **Restriction polymorphism** – Specialized enzymes called restriction endonucleases are used to cut the genome at specialized sites called restriction endonuclease recognition site and maps are prepared based on it.
- **Microsatellites** – these are repetitive DNA sequences.

### Salient features of HGP

- Human genome contains  $3 \times 10^9$  (3164.7 million) nucleotide bases.
- An average gene consists of 3000 bases. However, the size of genes varies. Largest known human gene is dystrophin (2.4 million bases).
- Total number of genes in human genome – 30,000
- Over 50% of the discovered genes have unknown functions.
- Less than 2% of genome codes for proteins.
- Large portion of genome consists of repeating sequences.
- Repetitive sequences have no coding function. They are repeated over hundred to thousand times. They may have a role in evolution, chromosome structure, and dynamics.
- Chromosome with most genes – Chromosome 1 (2968)  
Chromosome with fewest genes – Chromosomes Y (231)
- **SNPs** (single nucleotide polymorphism, pronounced as ‘snips’) occur at about 1.4 million locations in human DNA. They are believed to have significance in explaining diseases and tracing evolutionary history of human beings.

### DNA Fingerprinting

#### Introduction

- DNA fingerprinting is a method for comparing the DNA sequences of any two individuals.
- 99.9% of the base sequences in all human beings are identical. It is the remaining 0.1% that makes every individual unique.
- It is a really difficult and time-consuming task to sequence and compare all  $3 \times 10^9$  bases in two individuals. So, instead of considering the entire genome, certain specific regions called **repetitive DNA** sequences are used for comparative study.

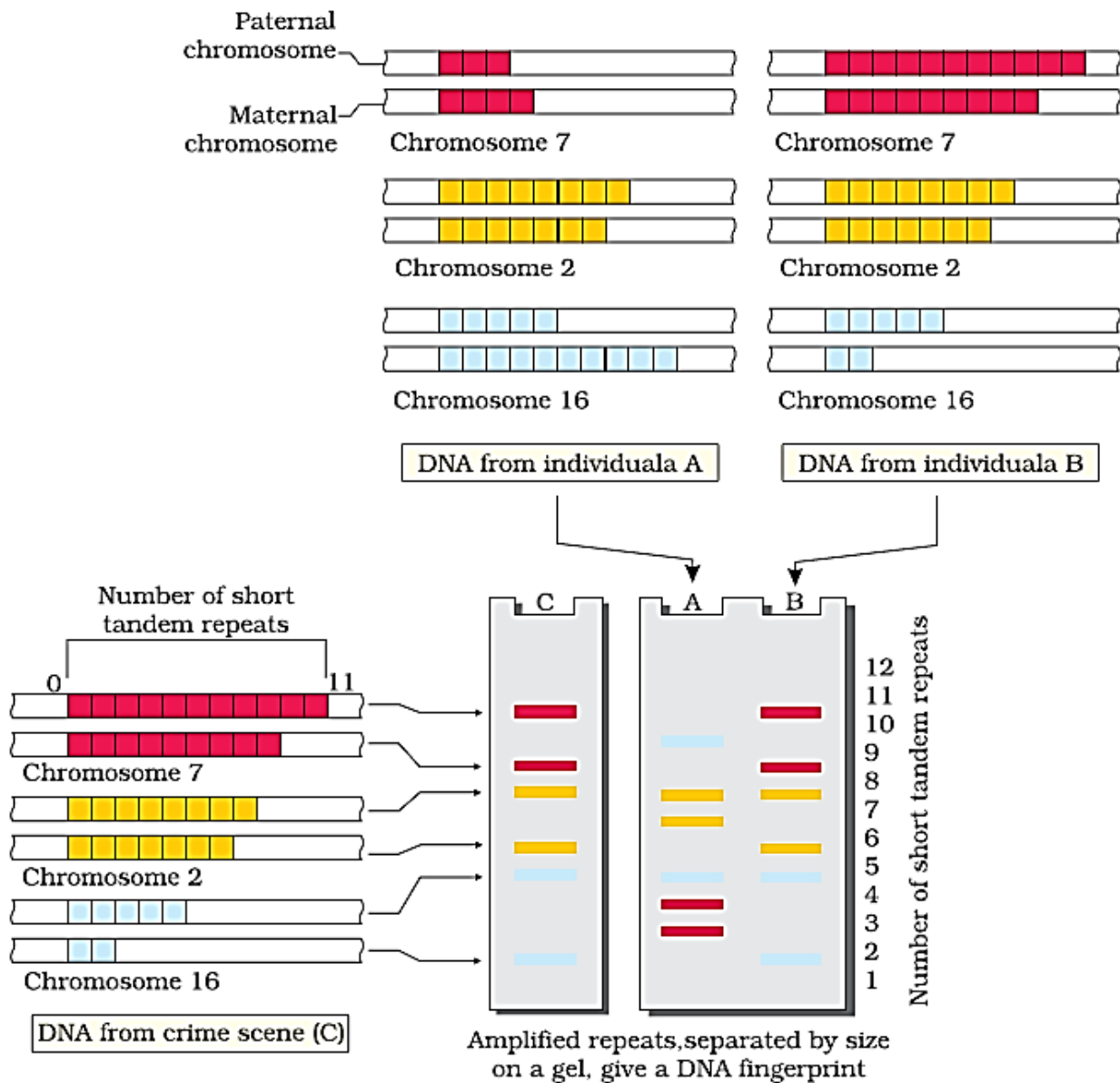
#### Basis of DNA Fingerprinting

- Repetitive DNA is separated from bulk genomic DNA since it appears as a distinct peak during density gradient centrifugation.
- Major peak: Formed by bulk DNA. Smaller peak: **Satellite DNA**
- Satellites are of two types—**micro-satellites** and **mini satellites**, depending upon the base composition, length of segment and the number of repetitive units.
- Satellites do not code for proteins, but have a major role to play in DNA fingerprinting.
- Polymorphism is actually a result of mutation. A germ cell mutation (which can pass on to the next generation through sexual reproduction) gives rise to polymorphism in populations.
- In other words, an inheritable mutation if observed in higher frequencies in a population is known as **polymorphism**.
- Polymorphisms arise normally in non-coding sequences because mutations in non-coding sequences do not affect an individual's reproductive ability.

#### Methodology of DNA fingerprinting

- DNA fingerprinting was initially developed by **Alec Jeffreys**.
- **VNTR (variable number of tandem repeats)** are satellite DNAs that show high degree of polymorphism.
- VNTRs are used as probes in DNA fingerprinting.
- First of all, DNA from an individual is isolated and cut with restriction endonucleases.
- Fragments are separated according to their size and molecular weight on gel electrophoresis.
- Fragments separated on electrophoresis gel are blotted (immobilised) by Southern blot on a synthetic membrane such as nylon or nitrocellulose.
- Immobilised fragments are hybridised with a VNTR probe.
- Hybridised DNA fragments can be detected by autoradiography.

- VNTRs vary in size from 0.1 to 20 kb.
- Hence, in the autoradiogram, band of different sizes will be obtained.
- These bands are characteristic for an individual. They are different in each individual, except identical twins.



**Figure:** Schematic representation of DNA fingerprinting: Few representative chromosomes have been shown to contain different copy number of VNTR. For the sake of understanding different colour schemes have been used to trace the origin of each band in the gel. The two alleles (paternal and maternal) of a chromosome also contain different copy numbers of VNTR. It is clear that the banding pattern of DNA from crime scene matches with individual B, and not with A.

### Applications of DNA Fingerprinting

- DNA fingerprinting is widely used in forensics since every DNA of every tissue from an individual has the same degree of polymorphism.
- DNA fingerprinting forms the basis of paternity testing since a child inherits polymorphism from both its parents.
- It can be used for studying genetic diversity in a population and evolution.

## SUMMARY

Nucleic acids are long polymers of nucleotides. While DNA stores genetic information, RNA mostly helps in transfer and expression of information. Though DNA and RNA both function as genetic material, but DNA being chemically and structurally more stable is a better genetic material. However, RNA is the first to evolve and DNA was derived from RNA. The hallmark of the double stranded helical structure of DNA is the hydrogen bonding between the bases from opposite strands. The rule is that Adenine pairs with Thymine through two H-bonds, and Guanine with Cytosine through three H-bonds. This makes one strand complementary to the other. The DNA replicates semi conservatively, the process is guided by the complementary H-bonding. A segment of DNA that codes for RNA may in a simplistic term can be referred as gene. During transcription also, one of the strands of DNA acts a template to direct the synthesis of complementary RNA. In bacteria, the transcribed mRNA is functional, hence can directly be translated. In eukaryotes, the gene is split. The coding sequences, exons, are interrupted by non-coding sequences, introns. Introns are removed and exons are joined to produce functional RNA by splicing. The messenger RNA contains the base sequences that are read in a combination of three (to make triplet genetic code) to code for an amino acid. The genetic code is read again on the principle of complementarity by tRNA that acts as an adapter molecule. There are specific tRNAs for every amino acid. The tRNA binds to specific amino acid at one end and pairs through H-bonding with codes on mRNA through its anticodons. The site of translation (protein synthesis) is ribosomes, which bind to mRNA and provide platform for joining of amino acids. One of the rRNA acts as a catalyst for peptide bond formation, which is an example of RNA enzyme (ribozyme). Translation is a process that has evolved around RNA, indicating that life began around RNA. Since, transcription and translation are energetically very expensive processes, these have to be tightly regulated. Regulation of transcription is the primary step for regulation of gene expression. In bacteria, more than one gene is arranged together and regulated in units called as operons. *Lac* operon is the prototype operon in bacteria, which codes for genes responsible for metabolism of lactose. The operon is regulated by the amount of lactose in the medium where the bacteria are grown. Therefore, this regulation can also be viewed as regulation of enzyme synthesis by its substrate.

Human genome project was a mega project that aimed to sequence every base in human genome. This project has yielded much new information. Many new areas and avenues have opened up as a consequence of the project. DNA Fingerprinting is a technique to find out variations in individuals of a population at DNA level. It works on the principle of polymorphism in DNA sequences. It has immense applications in the field of forensic science, genetic biodiversity and evolutionary biology.